

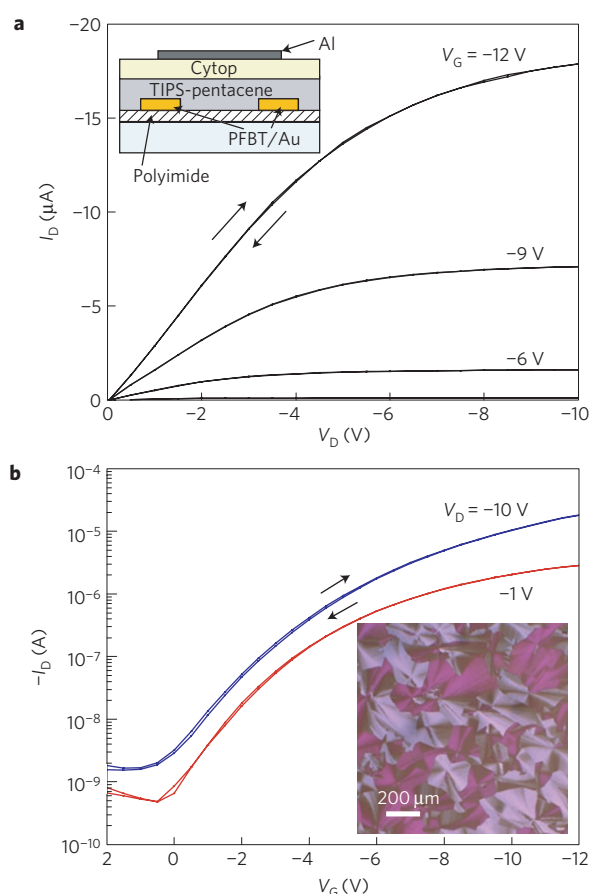
# Band-like temperature dependence of mobility in a solution-processed organic semiconductor

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The mobility  $\mu$  of solution-processed organic semiconductors has improved markedly<sup>1,2</sup> to room-temperature values of  $1\text{--}5\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ . In spite of their growing technological importance<sup>3</sup>, the fundamental open question remains whether charges are localized onto individual molecules or exhibit extended-state band conduction like those in inorganic semiconductors<sup>4</sup>. The high bulk mobility of  $100\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$  at 10 K of some molecular single crystals<sup>5</sup> provides clear evidence that extended-state conduction is possible in van-der-Waals-bonded solids at low temperatures. However, the nature of conduction at room temperature with mobilities close to the Ioffe–Regel limit remains controversial<sup>6</sup>. Here we investigate the origin of an apparent ‘band-like’, negative temperature coefficient of the mobility ( $d\mu/dT < 0$ ) in spin-coated films of 6,13-bis(triisopropylsilyl)ethynyl-pentacene. We use optical spectroscopy of gate-induced charge carriers to show that, at low temperature and small lateral electric field, charges become localized onto individual molecules in shallow trap states, but that a moderate lateral electric field is able to detrapp them resulting in highly nonlinear, low-temperature transport. The negative temperature coefficient of the mobility at high fields is not due to extended-state conduction but to localized transport limited by thermal lattice fluctuations.

Regarding the nature of charge transport at room temperature, some authors have suggested that the strong coupling to thermally excited intra- and intermolecular vibrations localizes the charge carriers<sup>7–9</sup>, whereas others have observed evidence of extended-state conduction even at room temperature<sup>10,11</sup>. On the surface of thin organic films in field-effect transistor (FET) configurations, a negative temperature coefficient ( $d\mu/dT < 0$ ), which is commonly considered as a fingerprint of band-like transport, has not yet been observed<sup>12,13</sup>. In most organic thin films, the field-effect mobility exhibits thermal activation behaviour determined by pronounced energetic disorder at the interface with the gate dielectric<sup>14</sup>. Recently, several high-mobility, semicrystalline conjugated polymers have been reported to exhibit highly nonlinear transport properties at low temperatures<sup>15,16</sup>. These have been interpreted as a manifestation of one-dimensional Luttinger liquid physics<sup>17</sup>, but their microscopic origin remains under debate<sup>18,19</sup>.

To investigate surface transport in highly ordered thin organic films over a wide temperature range we selected spin-coated 6,13-bis(triisopropylsilyl)ethynyl (TIPS)-pentacene films in contact with a perfluorinated, low-dielectric-constant polymer gate dielectric in a top-gate, bottom-contact FET architecture (inset of Fig. 1a). TIPS-pentacene formed uniform, polycrystalline films with a large domain size of over  $100\text{ }\mu\text{m}$  (inset of Fig. 1b). The devices exhibited *p*-channel FET characteristics with high room-temperature saturation and linear mobilities of  $1.2\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$  and  $0.8\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ , respectively, and negligible hysteresis in both output (Fig. 1a) and transfer characteristics (Fig. 1b).

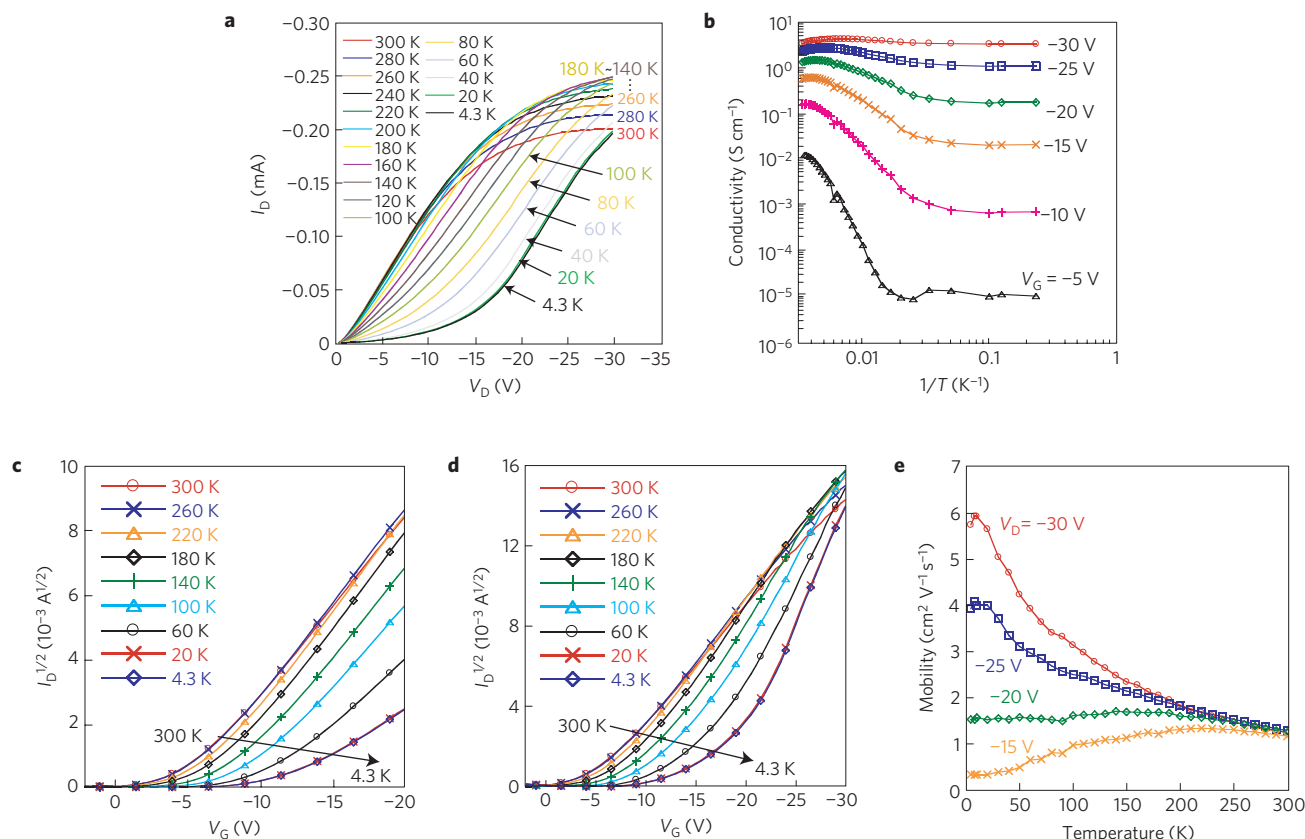


**Figure 1 | Characteristics of a TIPS-pentacene top-gate transistor.**

**a**, Output characteristics of a device with channel length  $L = 5\text{ }\mu\text{m}$ , channel width  $W = 100\text{ }\mu\text{m}$  and a 120-nm-thick dielectric. The inset shows a schematic of the device structure. PFBT: pentafluorobenzene thiol.

**b**, Transfer characteristics. The inset shows a polarized optical micrograph of a TIPS-pentacene film.

In devices with a relatively short channel length ( $L = 5\text{ }\mu\text{m}$ ), the temperature dependence of the transistor current varied strongly with applied electric field. For intermediate drain and gate voltages,  $V_D$  and  $V_G$ , respectively ( $V_D = V_G = -15\text{ V}$ ), the FET current was nearly temperature independent between room temperature and 200 K, but then decreased monotonously with decreasing temperature. At higher voltages ( $V_D = V_G = -30\text{ V}$ ), the FET current increased by  $\sim 25\%$  on cooling from room temperature to 140 K, before dropping slightly at even lower temperatures. However, even at 4.3 K the current remained about the same as

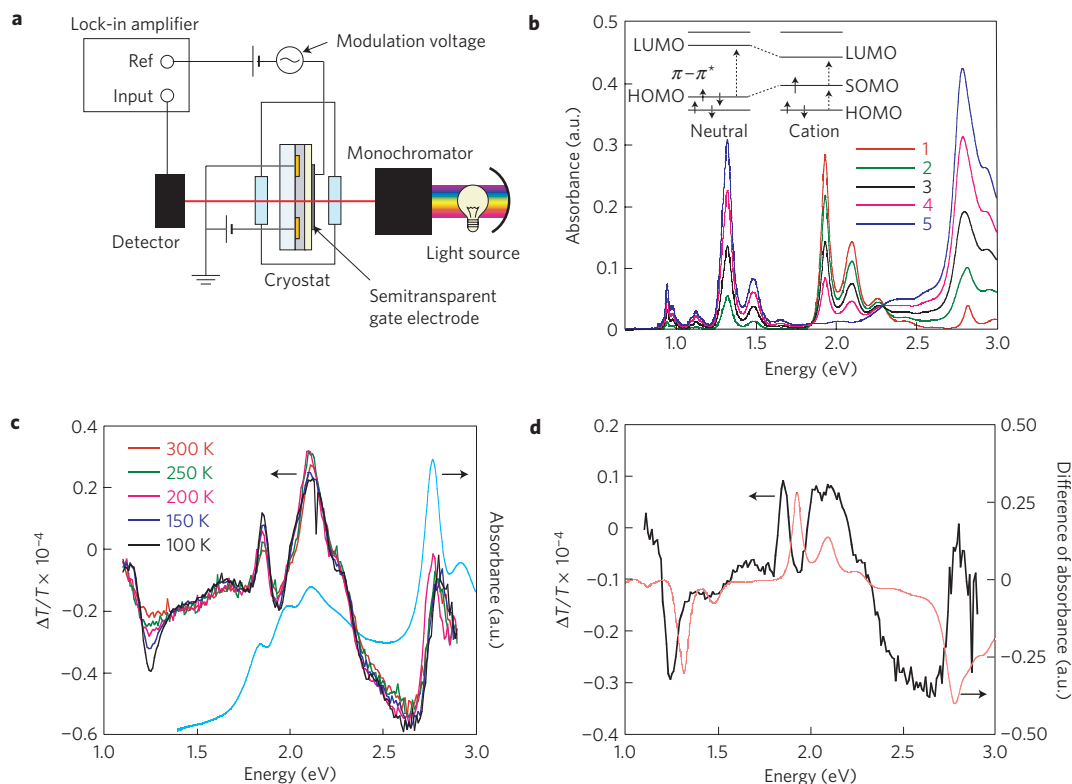


**Figure 2 | Temperature-dependent characteristics of TIPS-pentacene FETs with  $L = 5\text{ }\mu\text{m}$ ,  $W = 100\text{ }\mu\text{m}$  and a 120-nm-thick dielectric. a**, Temperature dependence of drain current  $I_D$  as a function of  $V_D$  at fixed  $V_G = -30$  V. **b**, Temperature dependence of the conductivity defined as the ratio of  $I_D/V_D$  at fixed  $V_D = -30$  V. Following refs 15–17, we calculated the conductivity by assuming the thickness of the accumulation layer to be 1 nm and dividing  $I_D$  by  $V_D$ . **c,d**, Square root of drain current as a function of gate voltage, with drain voltages of  $V_D = -15$  V (**c**) and  $V_D = -30$  V (**d**). **e**, Temperature dependence of the effective mobility of TIPS-pentacene FETs with different  $V_D$ .

at room temperature (Fig. 2a). The conductivity at high  $V_D$  and  $V_G$  is nearly temperature independent (Fig. 2b), which is very similar to the behaviour reported recently in some semiconducting polymers<sup>15–17</sup>. Down to about 140 K the output characteristics retained a near-textbook-like shape, with clearly defined linear and saturation regions. However, below 140 K they acquired a distinctly positive curvature at intermediate drain voltages (Fig. 2a). Such nonlinear transport properties were observed only when high drain and gate voltages up to  $\sim -30$  V were applied to short-channel devices. When voltages were limited to  $-15$  V (Supplementary Fig. S1) or the channel length was long ( $L = 20\text{ }\mu\text{m}$ ), the output characteristics showed clear saturation behaviour even at 4.3 K.

The increase of transistor current between room temperature and 140 K demonstrates unambiguously that, in contrast to other organic FETs, charge transport in TIPS-pentacene FETs at sufficiently high applied voltages improves with decreasing temperature. It is interesting to interpret this behaviour in terms of the temperature dependence of the field-effect mobility, which can be extracted from the slope of the square root of the drain current in the saturation regime<sup>20</sup> (Fig. 2c,d). This standard extraction method can be applied safely to the intermediate-voltage characteristics ( $|V_{D/G}| < 15$  V) at all temperatures, and to the high-voltage characteristics ( $|V_{D/G}| < 30$  V) down to about 140 K as the characteristics retain well-defined saturation regions. With decreasing temperature the square root of the saturated drain current exhibits a positive curvature at low gate voltages and the threshold voltage shifts to more negative values. This is commonly observed in organic FETs and reflects the gate voltage filling up low-mobility trap states induced by residual disorder,

and eventually populating more mobile states higher up in the interfacial density of states. The slope of the square root of the drain current at high gate voltages can be used to estimate the effective mobility of charges in these more mobile states. For an intermediate drain voltage,  $V_D = -15$  V, we observe a slight increase of this effective mobility with decreasing temperature in the temperature range 200 K  $< T < 300$  K (Fig. 2d), but for  $T < 200$  K the effective mobility becomes thermally activated with an activation energy,  $E_a = 5.7$  meV. This small value is consistent with the mobility becoming determined by a shallow trap state, and suggests that the degree of energetic disorder at the TIPS-pentacene/Cytop interface is significantly lower than in other solution-processed organic FETs (refs 21,22). For higher lateral electric fields ( $V_D = -30$  V), the slope of the saturated transfer characteristics at high gate voltages increases monotonically with decreasing temperature (Fig. 2e, Supplementary Fig. S2), and the extracted effective mobility reaches a value of  $2.5\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$  at 140 K (Fig. 2e). This negative temperature coefficient of the mobility is not an artefact of the extraction method. Clearly, to explain the increase of current down to 140 K, the effective mobility has to increase. In fact, because the mobility is evidently gate-voltage dependent, the effective mobility of charge states populated at higher gate voltages needs to increase even more strongly with decreasing temperature than the absolute current to compensate for the reduced contribution to the current from charges in lower-mobility trap states populated at low gate voltages. We also used an alternative transconductance method for extracting mobilities (Supplementary Fig. S3; ref. 20), and similar behaviour was obtained. We checked the potential influence of contact resistance,



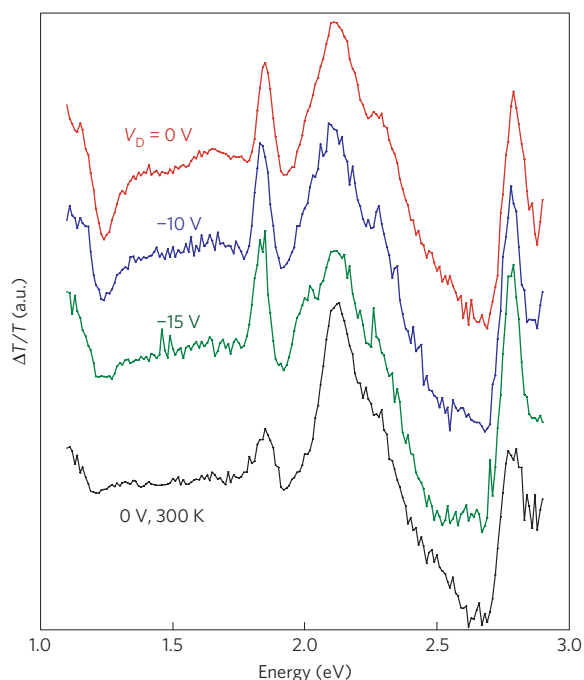
**Figure 3 | Optical spectroscopy of charge carriers in TIPS-pentacene.** **a**, Schematic of the experimental set-up for CMS measurements. **b**, Chemical doping absorption spectra of isolated, neutral and radical-cation TIPS-pentacene molecules in dilute solution. Spectrum 1 corresponds to the neutral-molecule absorption in undoped solution. Spectra 2–4 correspond to increasing levels of chemical doping. In spectrum 5 all molecules in solution have been doped. The close correspondence between the solution spectrum and the CMS data that are obtained at very low charge concentrations, of the order of  $10^{12} \text{ cm}^{-2}$ , makes it likely that the main charged species observed is a radical cation as opposed to a dication. The inset shows an energy diagram with the expected optical transitions for the neutral molecule and the radical cation. **c**, Temperature-dependent CMS spectra of TIPS-pentacene FETs ( $L = 40 \mu\text{m}$ , 450-nm-thick dielectric). The thin-film, ultraviolet-visible absorption spectrum of a TIPS-pentacene film prepared on a polyimide-coated glass substrate is also shown for comparison (light blue). At room temperature the optical cross-sections of the CMS charge-induced absorption are  $1.6 \times 10^{-16}$  and  $6.7 \times 10^{-17} \text{ cm}^2$  at 1.24 and 2.6 eV, respectively. They are of the same order of magnitude as those determined from the chemical doping spectra. **d**, Comparison of the CMS spectrum at 43 K (black) with the difference between the absorption spectra of a chemically doped and an undoped solution (red).

which manifests itself in the room-temperature characteristics as a slight suppression of the current at low drain voltages ( $|V_D| < 5 \text{ V}$ ). By comparing four-point and two-point probe measurements, we found that contact-corrected and uncorrected mobilities exhibit very similar temperature dependence with the same activation energy (Supplementary Fig. S4). Below 140 K one might argue that the method of extracting effective mobilities from the saturated transfer characteristics at  $V_D = -30 \text{ V}$  is not reliable because the characteristics no longer exhibit a well-defined saturation regime. However, even at such low temperature we can still see indications that the device would enter saturation if application of even higher drain voltages was not prevented by dielectric breakdown (Fig. 2a and Supplementary Fig. S1). The effective mobility values extracted in this regime of, for example,  $6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$  at 4.3 K, are therefore likely to be an underestimate of the real mobility values, because the true saturated current values are higher than the currents reached at  $V_D = -30 \text{ V}$ .

These electrical characteristics, particularly the pronounced nonlinear lateral-field dependence of the low-temperature transport, are reminiscent of those observed recently in polythiophene-based FETs (refs 15–17), which were interpreted as a manifestation of Luttinger-liquid physics<sup>17</sup>. For our TIPS-pentacene system we propose a different explanation based on optical charge modulation spectroscopy (CMS). Localized, polaronic charges in organic semiconductors have characteristic optical absorptions differing

from the neutral  $\pi-\pi^*$  absorption, such as the transitions between the highest occupied molecular orbital (HOMO) and the singly occupied molecular orbital (SOMO), and between the SOMO and the lowest unoccupied molecular orbital<sup>23,24</sup> (LUMO) of the radical cation (see inset of Fig. 3b). CMS detects these by measuring changes  $\Delta T$  of the optical transmission  $T$  through the semitransparent FET on modulation of the gate voltage/carrier concentration<sup>21,25</sup> (Fig. 3a). CMS provides us with a direct, spectroscopic probe to prove that the electrical characteristics reflect real changes in the molecular nature of the charge carriers in the accumulation layer as a function of temperature and electric field, and are not due to artefacts caused by the method of mobility extraction or by contact effects.

We first carried out spectroscopy of isolated charged molecules in dilute solution (Fig. 3b). When adding successive amounts of the oxidative dopant  $\text{FeCl}_3$  to a solution of neutral TIPS-pentacene, the characteristic HOMO–LUMO absorption at around 1.8–2.5 eV was bleached, and characteristic charge-induced absorptions appeared at 0.9–1.7 eV with a pronounced vibronic structure. The peak at 1.32 eV with a vibronic replica at 1.48 eV can be attributed to the HOMO–SOMO transition, whereas the peak at 0.94 eV could be due to the SOMO–LUMO transition. We also detect a higher-lying charge-induced absorption around 2.8 eV. The isolated radical-cation spectrum in dilute solution provides a reference for interpreting the CMS data obtained in the crystalline, solid state.



**Figure 4 | Drain-voltage dependence of the CMS spectra at 100 K of a TIPS-pentacene FET with  $L = 5 \mu\text{m}$  and a 250-nm-thick dielectric.** For comparison, the corresponding spectrum at 300 K and  $V_D = 0$  V is also shown. All spectra were acquired with a gate voltage of  $-20$  V and a modulation bias of  $\pm 2$  V at 37 Hz.

The CMS spectra of TIPS-pentacene FETs taken at zero lateral electric field exhibit a characteristic bleaching region around 1.8–2.2 eV ( $\Delta T/T > 0$ ), owing to the reduction of the number of neutral molecules on charge injection (Fig. 3c). Two of the vibronic peaks observed in the thin-film absorption spectrum at 1.8 and 2.11 eV are also observed in the CMS spectrum (the third peak at 1.96 eV is visible in CMS as a weak shoulder). The shape of the bleaching signal is different because it is superimposed by a broad charge-induced absorption. We detect two charge-induced absorption regions ( $\Delta T/T < 0$ ) with peaks around 1.2 eV and 2.7 eV. The latter, which corresponds to the high-energy radical-cation transition in the chemical doping spectra, does not show much temperature dependence. However, the lower-energy feature exhibits an interesting temperature dependence. Near room temperature, the charge-induced absorption between 1.2–1.9 eV is broad and featureless without apparent vibronic structure. However, below 150–200 K the peak around 1.24 eV sharpens considerably and develops a vibronic replica around 1.45 eV. At low temperatures, the spectral shape becomes very similar to that of the isolated radical cation in solution although the peak position in the solid state is shifted by 80 meV to lower energies (Fig. 3d), possibly owing to differences in polarization between the solution and solid state.

The temperature at which this sharpening of the CMS spectrum is observed corresponds well to the temperature at which the mobility at intermediate lateral electric fields crosses over from a thermally activated to a temperature-independent/band-like regime. We conclude that the thermal activation of the mobility is due to trapping of charges in shallow traps in the organic semiconductor, which localize the charges onto single TIPS-pentacene molecules. We can exclude trapping on chemically modified TIPS-pentacene molecules, other chemical impurities or states in the gate dielectric, because one would then expect the CMS spectrum of the trapped charge to differ significantly from that of the isolated radical cation. The shallow trap states

could in principle be associated with misaligned molecules at grain boundaries. However, we consider it more likely that they are associated with structural defects within the grains, such as static molecular sliding defects at the active interface<sup>26</sup>, because our channel length is significantly shorter than typical crystalline domain sizes (inset of Fig. 1b), and the activation energy is small and similar to the one measured in single-grain pentacene FETs (ref. 27).

We can also draw important conclusions about charge transport at room temperature. The mere observation of a pronounced charge-induced absorption resembling that of the radical cation in solution suggests that, in spite of the negative temperature coefficient of the effective mobility, the charges are not fully extended, but remain localized over a certain number  $N$  of molecules.  $N$  is not necessarily equal to one as for the shallow trap state occupied at low temperatures, because the room-temperature CMS spectrum is much broader than the radical-cation solution spectrum, and lacks vibronic structure. However,  $N$  can not be macroscopic either because each individual molecule carrying a fractional charge  $e/N$  would then be expected to exhibit an optical absorption similar to that of the neutral molecule, and only a broad Drude-like optical response would be expected in CMS (refs 10, 28). This is clearly inconsistent with the room-temperature CMS spectrum, particularly the pronounced high-energy charge-induced absorption at 2.8 eV. For vacuum-sublimed pentacene FETs  $N$  was estimated to be  $\sim 10$  from electron-spin resonance experiments<sup>29</sup>.

The simultaneous observation of a negative temperature coefficient of the effective mobility and a pronounced charge-induced absorption of the carriers is consistent with the localization of the charge carriers being brought about, not by polaron self-localization, but by dynamic, intermolecular lattice disorder<sup>7</sup>. At any time, charges can be considered to be effectively localized in a region bound by sites where unfavourable configurations for intermolecular charge transfer are encountered. As CMS effectively averages over all such charge configurations in the accumulation layer, this could explain the broad, featureless room-temperature charge-induced absorption. In such dynamic-disorder-limited transport the temperature dependence of the mobility arises from freezing of low-energy intermolecular vibrations<sup>7</sup>.

Finally, we discuss the origin of the nonlinear dependence of the low-temperature transport on the lateral electric field. We have carried out low-temperature CMS experiments with a constant lateral electric field applied along the FET channel (Fig. 4). The sharp, trapped-charge absorption peak at 1.24 eV ( $V_D = 0$  V) becomes broadened with increasing lateral field. The CMS spectrum at 100 K and  $V_D = -10$  V resembles the one taken at 150 K without applied drain voltage, and the spectrum at 100 K and  $V_D = -15$  V is very similar to the broad, featureless charge-induced absorption observed at 300 K without drain voltage. This provides clear spectroscopic evidence that charges in single-molecule, shallow trap states can be effectively detrapped into more mobile states by application of the source–drain electric field. It is this process that is responsible for the nonlinear transport properties observed in TIPS-pentacene FETs at low temperature. Our results are consistent with previous models of temperature-independent field-emission tunnelling at low temperature<sup>18,19</sup>. We have fitted the electric-field dependence of the temperature-independent effective mobility in the range 4–20 K (Fig. 2c) to a simple Fowler–Nordheim tunnelling expression through a triangular barrier, and have extracted barrier heights/trap depths of 13–17 meV (Supplementary Fig. S5). Such small values are consistent with the trap depth estimated from the activation energy of the low-field mobility.

## Methods

To investigate surface transport in highly ordered thin organic films over a wide temperature range we selected TIPS-pentacene in contact with a perfluorinated, low-dielectric-constant Cytop polymer gate dielectric. We use a top-gate, bottom-contact device architecture (inset of Fig. 1a) with spin-coated films of

TIPS-pentacene and Cytop. A polyimide layer was inserted to control the surface wettability of the glass substrate. On polyimide, TIPS-pentacene formed uniform, polycrystalline films with a large domain size of over 100  $\mu\text{m}$  (inset of Fig. 1b).

A 50-nm-thick precursor of polyimide (PI-2525, HD Microsystems) was spin-coated onto the cleaned glass substrates and cured in a nitrogen atmosphere at 160 °C for 1 h, followed by 300 °C for 3 h. Standard lift-off photolithography was carried out on the polyimide-coated glass to define the 10-nm-thick Au source and drain electrodes. The Au patterned substrates were cleaned by acetone and isopropyl alcohol with ultrasonication, and treated with oxygen plasma at 150 W for 1 min. The gold contacts were treated with a 10 mM solution of pentafluorobenzene thiol in isopropyl alcohol for 2 min to reduce contact resistance. The TIPS-pentacene and Cytop layers were prepared by spin-coating in a nitrogen atmosphere. A 10 mg ml<sup>-1</sup> solution of TIPS-pentacene in tetralin was spun at 1,000 r.p.m. for 1 min, followed by drying on a hotplate at 100 °C for 5 min. A layer of Cytop (obtained from Asahi Glass Co.) was successively formed on the TIPS-pentacene film and dried at 90 °C for 20 min. For measurements at high lateral electric field/source-drain voltage in devices with a short,  $L = 5 \mu\text{m}$ , channel length we reduced the thickness  $d$  of the gate dielectric to 120–250 nm to ensure correct device scaling and obtain clean saturation characteristics at room temperature. For longer-channel,  $L = 40 \mu\text{m}$ , devices we used  $d = 450 \text{ nm}$ . The CMS measurements in Fig. 3 were acquired in accumulation mode with a d.c. gate voltage of  $-35 \text{ V}$  and an a.c. modulation bias of  $\pm 5 \text{ V}$  at 37 Hz without drain voltage ( $V_D = 0$ ). The capacitances of the Cytop layers were  $28 \text{ nF cm}^{-2}$ ,  $14 \text{ nF cm}^{-2}$  and  $5.2 \text{ nF cm}^{-2}$  for 120-nm-, 250-nm- and 450-nm-thick films, respectively. These were determined by measuring metal-insulator-metal diode structures of Al/Cytop/Al. For the gate electrode, a 6–20-nm-thick aluminium electrode was prepared by vacuum evaporation through a shadow mask.

The chemical doping was carried out by adding a concentrated ( $1 \times 10^{-3} \text{ M}$ )  $\text{FeCl}_3$  chloroform solution into the dilute ( $1 \times 10^{-5} \text{ M}$ ) TIPS-pentacene chloroform solution.  $\text{FeCl}_3$  solution (140  $\mu\text{l}$ ) was added to 2 ml of TIPS-pentacene solution to obtain a fully oxidized spectrum (5 in Fig. 3b). Some unreacted  $\text{FeCl}_3$  molecules remain in solution.

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## Author contributions

T.S. carried out the experiments. T.S. and H.S. developed the interpretation of the data and wrote the manuscript.

## Additional information

The authors declare no competing financial interests. Supplementary information accompanies this paper on [www.nature.com/naturematerials](http://www.nature.com/naturematerials). Reprints and permissions information is available online at <http://npg.nature.com/reprintsandpermissions>. Correspondence and requests for materials should be addressed to T.S. or H.S.